Harsh Parikh

Assistant Professor · Department of Biostatistics · Yale University

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SUMMARY

I develop machine learning–aided causal inference approaches to solve high-stakes problems that are: (i) Accurate, enabling estimation of heterogeneous treatment effects in complex scenarios with limited data; (ii) Trustworthy, allowing domain experts to understand the machinery, validate underlying assumptions, and identify where predictions may be unreliable; and (iii) Domain-conscious, leveraging domain context and knowledge to come up with applicable solutions, reducing the research-to-practice gap.

CURRENT AFFILIATIONS

2025–Present
Tenure-Track Assistant Professor
Department of Biostatistics, Yale School of Public Health

2025–Present
Assistant Professor (Secondary Appointment)
Department of Statistics and Data Science, Yale University

2025–Present
Affiliated Faculty
Foundations of Data Science, Yale University

2025–Present
Affiliated Faculty
Institution for Social and Policy Studies (ISPS), Yale University

2025–Present
Guest Researcher
Danish Centre for Health Economics, Syddansk Universitet

2025–Present
Affiliate Researcher
Johns Hopkins Bloomberg School of Public Health

2025–Present
Applied Scientist III
Amazon.com · Supply Chain Optimization Technologies

ACADEMIC TRAINING

2023–2025
Postdoctoral Fellow
Johns Hopkins Bloomberg School of Public Health, Department of Biostatistics
Trustworthy Causal Inference for Transportability and Generalizability
Advisors: Elizabeth Stuart, Kara Rudolph

2018–2023

Ph.D. in Computer Science

Duke University, Department of Computer Science

Causal Inference for High-Stakes Decisions

Advisors: Cynthia Rudin, Alexander Volfovsky, Sudeepa Roy

★ Outstanding PhD Dissertation Award 2023 · Certificate in College Teaching

2016–2018

M.S. Economics & Computation

Duke University, Department of Economics

Advisors: Vincent Conitzer, Charles Becker

2011–2015

B.Tech. Computer Science & Engineering

Indian Institute of Technology (IIT) Delhi

Advisor: Parag Singla

PUBLICATIONS

36. Yilin Song, Dan M Kluger, **Harsh Parikh**, Tian Gu. Demystifying Prediction Powered Inference, 2026 `2 cited` `co-senior/corresponding author`
35. David Arbour*, **Harsh Parikh***, Bijan Niknam, Elizabeth Stuart, Kara Rudolph, Avi Feller. Regularizing Extrapolation in Causal Inference. *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2026 `2 cited` `co-first author`
34. Emily K Johnson, **Harsh Parikh**, Kim Rose Olsen, Angela Y Chang, Liza Sopina. Lasting Income Costs of Mental and Physical Illness, 2026
33. Camille DeSisto, Ranaivo Rasolofoson, Michelle Foley, **Harsh Parikh**. When Does Agroforestry Income Reduce Deforestation? Evidence from a Natural Experiment in Madagascar, 2026
32. **Harsh Parikh**, Gabriel Levin-Konigsberg, Dominique Perrault-Joncas, Alexander Volfovsky. TEA-Time: Transporting Effects Across Time, 2026
31. Qi Zhang, **Harsh Parikh**, Ashley Naimi, Razieh Nabi, Christopher Kim, Timothy Lash. Controllable Generative Sandbox for Causal Inference, 2026
30. **Harsh Parikh**. Why are there many equally good models? An Anatomy of the Rashomon Effect, 2026
29. Yiran Wang, Alicia E Boyd, Lillian Rountree, Yi Ren, Kate Nyhan, Ruchit Nagar, Jackson Higginbottom, Megan L Ranney, **Harsh Parikh**, Bhramar Mukherjee. Ten Core Concepts for Ensuring Data Equity in Public Health. *JAMA Health Forum*, 2026
28. **Harsh Parikh**, Rachael K Ross, Elizabeth Stuart, Kara E Rudolph. Who Are We Missing?: A Principled Approach to Characterizing the Underrepresented Population. *Journal of the American Statistical Association*, 2025 `13 cited`
27. **Harsh Parikh**, Marco Morucci, Vittorio Orlandi, Sudeepa Roy, Cynthia Rudin, Alexander Volfovsky. A Double Machine Learning Approach for Combining Experimental and Observational Studies. *Observational Studies*, 2025 `10 cited`
26. Seyedeh Baharan Khatami, **Harsh Parikh**, Haowei Chen, Sudeepa Roy, Babak Salimi. Graph Machine Learning based Doubly Robust Estimator for Network Causal Effects. *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2025 `4 cited`
25. Yiran Wang, Alicia E Boyd, Lillian Rountree, Yi Ren, Kate Nyhan, Ruchit Nagar, Jackson Higginbottom, Megan L Ranney, **Harsh Parikh**, Bhramar Mukherjee. Towards Enhancing Data Equity in Public Health Data Science. *arXiv preprint arXiv:2508.20301*, 2025 `3 cited`
24. **Harsh Parikh**, Trang Quynh Nguyen, Elizabeth A Stuart, Kara E Rudolph, Caleb H Miles. A Cautionary Tale on Integrating Studies with Disparate Outcome Measures for Causal Inference. *Advances in Neural Information Processing Systems (NeurIPS)*, 2025 `2 cited`
23. Carly L Brantner, Trang Quynh Nguyen, **Harsh Parikh**, Congwen Zhao, Hwanhee Hong, Elizabeth A Stuart. Precision mental health: predicting heterogeneous treatment effects for depression through data integration. *Journal of the Royal Statistical Society Series C: Applied Statistics*, 2025 `1 cited`

22. Grace V Ringlein, Trang Quynh Nguyen, Peter P Zandi, Elizabeth A Stuart, **Harsh Parikh**. Demystifying Proximal Causal Inference, 2025
21. Emily K Johnson, **Harsh Parikh**, Kim Rose Olsen, Angela Y Chang, Liza Sopina. Breast cancer and income loss in Denmark: heterogeneous outcomes and longitudinal effects. *Nature Communications*, 2025
20. CFPM de Sousa, **Harsh Parikh**, JD Bradley, Elizabeth Stuart, C Hu. Representativeness and Generalizability of NCI-Funded Multi-Modality Randomized Clinical Trials (RCTs): A Case Study of Locally-Advanced Non-Small Cell Lung Cancer (LA-NSCLC .... *International Journal of Radiation Oncology, Biology, Physics*, 2025
19. Quinn Lanners, Cynthia Rudin, Alexander Volfovsky, **Harsh Parikh**. Data Fusion for Partial Identification of Causal Effects. *Advances in Neural Information Processing Systems (NeurIPS)*, 2025
18. Srikar Katta, **Harsh Parikh**, Cynthia Rudin, Alexander Volfovsky. Interpretable causal inference for analyzing wearable, sensor, and distributional data. *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2024 `11 cited`
17. Melody Y Huang, **Harsh Parikh**. Towards generalizing inferences from trials to target populations. *Harvard Data Science Review*, 2024 `8 cited`
16. **Harsh Parikh**, Haoqi Sun, Rajesh Amerineni, Eric S Rosenthal, Alexander Volfovsky, Cynthia Rudin, M Brandon Westover, Sahar F Zafar. How many patients do you need? Investigating trial designs for anti-seizure treatment in acute brain injury patients. *Annals of Clinical and Translational Neurology*, 2024 `4 cited`
15. **Harsh Parikh**[*], Quinn M Lanners[*], Zade Akras, Sahar Zafar, M Brandon Westover, Cynthia Rudin, Alexander Volfovsky. Safe and interpretable estimation of optimal treatment regimes. *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2024 `4 cited` `co-first author`
14. **Harsh Parikh**[*], Kentaro Hoffman[*], Haoqi Sun[*], Sahar F Zafar, Wendong Ge, Jin Jing, Lin Liu, Jimeng Sun, Aaron Struck, Alexander Volfovsky, Cynthia Rudin, M Brandon Westover. Effects of epileptiform activity on discharge outcome in critically ill patients in the USA: a retrospective cross-sectional study. *The Lancet Digital Health*, 2023 `34 cited` `co-first author`
13. Quinn Lanners, **Harsh Parikh**, Alexander Volfovsky, Cynthia Rudin, David Page. Variable importance matching for causal inference. *Conference on Uncertainty in Artificial Intelligence (UAI)*, 2023 `9 cited`
12. **Harsh Parikh**. Synthetic Control as Balancing Scores. *International Conference on Learning Representations (ICLR), Tiny Papers*, 2023
11. **Harsh Parikh**, Carlos Varjao, Louise Xu, Eric Tchetgen Tchetgen. Validating causal inference methods. *International Conference on Machine Learning (ICML)*, 2022 `75 cited`
10. **Harsh Parikh**, Alexander Volfovsky, Cynthia Rudin. MALTS: Matching After Learning to Stretch. *Journal of Machine Learning Research*, 2022 `49 cited`
9. Haoning Jiang, Thomas Howell, Neha R Gupta, Vittorio Orlandi, Marco Morucci, **Harsh Parikh**, Sudeepa Roy, Cynthia Rudin, Alexander Volfovsky. AME: Interpretable Almost Exact Matching for Causal Inference. *Advances in Neural Information Processing Systems (NeurIPS), Demonstrations Track*, 2022
8. Amir Gilad, **Harsh Parikh**, Sudeepa Roy, Babak Salimi. Heterogeneous Treatment Effects in Social Networks, 2021 `3 cited`
7. Babak Salimi, **Harsh Parikh**, Moe Kayali, Lise Getoor, Sudeepa Roy, Dan Suciu. Causal relational learning, 2020 `65 cited`
6. Sarul Malik, **Harsh Parikh**, Neil Shah, Sneh Anand, Shalini Gupta. Non-invasive platform to estimate fasting blood glucose levels from salivary electrochemical parameters. *Healthcare Technology Letters*, 2019 `17 cited`
5. **Harsh Parikh**, Cynthia Rudin, Alexander Volfovsky. An Application of Matching After Learning To Stretch (MALTS). *Observational Studies*, 2019 `11 cited`
4. Shayoni Dutta, Spandan Madan, **Harsh Parikh**, Durai Sundar. An ensemble micro neural network approach for elucidating interactions between zinc finger proteins and their target DNA. *BMC Genomics*, 2016 `13 cited`
3. Sarul Malik, Shalini Gupta, **Harsh Parikh**, Sneh Anand. Gargling affect on salivary electrochemical parameters to predict blood glucose. *IEEE International Conference on Computational Techniques in Information and Communication Technologies (ICCTICT)*, 2016 `1 cited`
2. **Harsh Parikh**, Apoorvi Singh, Annangarachari Krishnamachari, Kushal Shah. Computational prediction of origin of replication in bacterial genomes using correlated entropy measure (CEM). *Biosystems*, 2015 `9 cited`

1. Alok Shankar Mysore, Vikas S Yaligar, Imanol Arrieta Ibarra, Camelia Simoiu, Sharad Goel, Ramesh Arvind, Chiraag Sumanth, Arvind Srikantan, Bhargav HS, Mayank Pahadia, Tushar Dobha, Atif Ahmed, Mani Shankar, Himani Agarwal, Rajat Agarwal, Sai Anirudh-Kondaveeti, Shashank Arun-Gokhale, Aayush Attri, Arpita Chandra, Yogitha Chilukur, Sharath Dharmaji, Deepak Garg, Naman Gupta, Paras Gupta, Glincy Mary Jacob, Siddharth Jain, Shashank Joshi, Tarun Khajuria, Sameeksha Khillan, Sandeep Konam, Praveen Kumar-Kolla, Sahil Loomba, Rachit Madan, Akshansh Maharaja, Vidit Mathur, Bharat Munshi, Mohammed Nawazish, Venkata Neehar-Kurukunda, Venkat Nirmal-Gavarraju, Sonali Parashar, **Harsh Parikh**, Avinash Paritala, Amit Patil, Rahul Phatak, Mandar Pradhan, Abhilasha Ravichander, Krishna Sangeeth, Sreecharan Sankaranarayanan, Vibhor Sehgal, Ashrith Sheshan, Suprajha Shibiraj, Aditya Singh, Anjali Singh, Prashant Sinha, Pushkin Soni, Bipin Thomas, Kasyap Varma-Dattada, Sukanya Venkataraman, Pulkit Verma, Ishan Yelurwar. Investigating the" wisdom of crowds" at scale, 2015 `6 cited`

Auto-updated from Google Scholar | 36 publications | 356 total citations

HONORS & AWARDS

2024	Future Leader in Data Science and AI, Michigan Institute for Data & AI in Society
2024	Selected for Building Future Faculty Program, North Carolina State University
2023	Outstanding PhD Dissertation Award, Dept. of Computer Science, Duke University
2023	Uncertainty in Artificial Intelligence Conference Travel Award
2022	Finalist, Two Sigma PhD Fellowship
2022	International Conference on Machine Learning Conference Travel Award
2022	Certificate in College Teaching, Duke University
2020-22	Amazon Graduate Research Fellowship
2020	Invited Talk, IIT Gandhinagar "Sabarmati Young Researcher's Seminar Series"
2016-18	Duke Economics Master's Scholar Award, Duke University
2016	Runner's Up, Global Healthcare Summit (Non-invasive blood glucose sensor)
2013-14	Charpak (Student Exchange) French Government Scholarship, University of Lorraine
2013	Summer Undergraduate Research Award (UROP), IIT Delhi
2011-12	IIT Delhi Semester Merit Award
2012	Runner's Up, CanSat USA by AAS, AIAA, JPL, NASA, NRL
2009-11	Manish Bhatt Scholarship, Excellence in Computer Science

ADDITIONAL PROFESSIONAL EXPERIENCE

May-Aug 2022

Research Intern

Meta (Facebook) · Core Data Science · New York

Inferring Network Interference in Randomized Controlled Trials

2020 & 2021

Applied Scientist Intern

Amazon.com · Seller Fees and Profitability · Seattle

Evaluating Causal Inference Methods

Jun-Jul 2017

Research Intern

The Urban Institute · International Development and Governance · Washington DC

Public Transport and Rental Markets; Women Empowerment and Labor Force Participation

Mar-Jul 2016

Research Fellow

Vision India Foundation · Evidence-based Public Policy Analysis · New Delhi

Impact Analysis of National Rural Employment Guarantee Act

Jul 2015–May 2016

Research Engineer

IBM India Research Laboratory · Data Fusion & Graph Analytics · New Delhi

Social Network Data Analysis for Law Enforcement

May–Jul 2014

Software Engineering Intern

Arista Networks · Emerging Technologies · Bangalore

Protocol for Audio–Video Bridging (AVB) Switches

INVITED & CONFERENCE TALKS

Causal Inference Beyond Positivity Assumption

IMSI Workshop: New Horizons on Model Transportability and Data Integration, Chicago (Jun 2026)

Transporting Effects Across Networks

Network Science 2026, CausNetS: Toward a Causal Network Science, Invited Speaker (2026)

Reinforcement Learning for Optimal Decisions in Public Health (Discussant)

ENAR 2026, Houston (Mar 2026)

Regularizing Extrapolation in Causal Inference

Yale Foundations of Data Science Colloquium, New Haven (Jan 2026) · Stats Seminar (Jan 2026) · IIT Gandhinagar (Jan 2026) · Indian Statistical Institute, Delhi (Jan 2026) · CMStats, London (Dec 2025) · NUS Singapore, IMS Young Mathematical Scientists Forum (Nov 2025)

Causal Inference Beyond Support

Yale Foundations of Data Science Colloquium, New Haven (Jan 2026)

Data Fusion for Partial Identification of Causal Effects

NeurIPS 2025, San Diego (Dec 2025)

A Cautionary Tale on Integrating Data with Disparate Outcomes

NeurIPS 2025, San Diego (Dec 2025)

Rashomon Set of Optimal Trees

Joint Statistical Meeting, Nashville (Aug 2025)

Machine Learning–Aided Causal Inference

University of Southern Denmark, DaCHE (2025)

Who Are We Missing? A Principled Approach to Identifying Underrepresented Groups

ACIC 2025, Detroit · ENAR 2025, New Orleans

Interpretable Machine Learning & Causal Inference for Advancing Healthcare and Public Health

Yale School of Public Health, New Haven (Jan 2025) · Johns Hopkins University, Applied Math & Statistics, Baltimore (Feb 2025) · Harvard University, Dept. of Statistics, Cambridge (Jan 2025) · Columbia University, Dept. of Biostatistics, New York (Jan 2025) · Boston University, Dept. of Biostatistics

(Nov 2024) · University of Michigan, Dept. of Biostatistics, Ann Arbor (Nov 2024) · UT Austin, Information, Risk, and Operations Management (Nov 2024)

Integrating Multiple Datasets with Disparate Outcomes for Efficient Causal Inference

INFORMS 2024, Seattle (Oct 2024)

Characterizing Underrepresented Populations when Generalizing Experimental Evidence

ICHPS (Jan 2025) · IIT Gandhinagar (Nov 2024) · ACIC (May 2024) · NC State University (Mar 2024) · ENAR (Feb 2024) · ICERM (Nov 2023)

A Double Machine Learning Approach to Combining Experimental and Observational Studies

INFORMS Annual Meeting (Oct 2023) · IISA Annual Meeting (Dec 2022)

Causal Inference for High-Stakes Decision Making

Wake Forest University School of Medicine (Mar 2023) · NC State University (Jan 2023) · MIDAS, Johns Hopkins (Sep 2022) · Jacobs Technion-Cornell Institute (Oct 2022) · Microsoft Research (Nov 2022)

Validating Causal Inference Methods

ICML (Jul 2022) · Clinical Data Animation Center, MGH (Aug 2022) · SER Conference (Jun 2023)

Matching After Learning to Stretch

ICML (Aug 2023) · Duke Microeconometrics (Sep 2019) · IIT Gandhinagar (Dec 2019)

Effect of Epileptiform Activity in Critically Ill Patients

Clinical Data Animation Center, MGH (Sep 2021)

TEACHING

Fall 2026	**Instructor — Introduction to Data Science** Yale University
2024	**Instructor — Interpretable Machine Learning Tutorial** International Conference on Computational Social Science (IC2S2)
Fall 2019	**Instructor — Introduction to Causal Inference (Advanced)** Duke Datathon
Spring 2019	**Teaching Assistant — COMPSCI 671 Machine Learning** Duke University
Fall 2018	**Instructor — Introduction to Data Science** Duke MEMPDC (Consulting Club)
Fall 2018	**Teaching Assistant — COMPSCI 590.02 Computational Microeconomics** Duke University

Spring 2018 — **Teaching Assistant — COMPSCI 223 Computational Microeconomics**
Duke University

2016–2017 — **Teaching Assistant — COMPSCI 230 Discrete Mathematics**
Duke University (Fall 2017, Spring 2017)

Fall 2016 — **Teaching Assistant — COMPSCI 201 Data Structures & Algorithms**
Duke University

POPULAR MEDIA

Harsh Parikh, Ankita Gupta, Subham. Covid-19: Mitigating the risk from reverse migration. *Ideas for India*, 2020

Harsh Parikh, Kumar Subham. Efficacy of India's Covid-19 response. *Center for Soft Power*, 2020

Ammar Malik, Harsh Parikh. Rents are driven by the quality of public services, not proximity to transit. *Urban Wire: International Development*, 2017

Fenohasina Rakotondrazaka Maret, Harsh Parikh, Rachel Wilder. Empowering women through international tourism. *Urban Wire: International Development*, 2017

Harsh Parikh. Book Review: The Indian Economy—A Macroeconomic Perspective. *ARTNeT UNESCAP*, 2017

SERVICE

Reviewer

ICML (2026) · AISTATS (2021, 2023–26) · NeurIPS (2021, 2025) · Annals of Applied Statistics (2026) · JASA (2025) · JRSS-A (2025) · JRSS-B (2025) · JRSS (2021) · JMLR (2024) · PNAS (2024) · IISE Transactions on Healthcare Systems Engineering (2025) · Nature Human Behaviour (2022) · Management Science (2021–22)

Leadership

Project Manager, NC Voucher Program Evaluation (2017) · Project Manager, Slum Development, AINA IIT Delhi (2011–15) · Committee Chair, CS Dept. Socials, Duke (2019–20) · President, Duke Indian Students Association (2019–21) · Treasurer, Duke Cricket Team (2017–20)

SKILLS & COURSEWORK

Coursework — Causal Inference, Machine Learning, Bayesian Statistics, Reinforcement Learning, Algorithms, Probability & Stochastic Processes, Linear Algebra, Real Analysis, Econometrics, Micro/Macroeconomics

Programming — Python, Java, C/C++, STATA, R, MATLAB, SQL, HTML, PHP, Perl, ArcGIS

REFERENCES

Cynthia Rudin
Computer Science, Duke University
cynthia.rudin@duke.edu

Elizabeth Stuart
Biostatistics, Johns Hopkins
estuart@jhu.edu

Alexander Volfovsky
Statistical Science, Duke University
alexander.volfovsky@duke.edu

Sudeepa Roy
Computer Science, Duke University
sudeepa@cs.duke.edu

Bhramar Mukherjee
Biostatistics, Yale University
bhramar.mukherjee@yale.edu

Kara Rudolph
Epidemiology, Columbia University
kr2854@cumc.columbia.edu